Notes: Cohen, Diether, and Malloy (JF, 2007)

Supply and Demand Shifts in the Shorting Market

Contents

1 Big picture

- **Question.** How does the stock-lending market reveal information about future stock returns, and is the relevant signal a demand shock for shorting, a supply shock of lendable shares, or simply a high shorting cost?
- **Core contribution.** The paper separates shifts in the supply of lendable shares from shifts in shorting demand by jointly observing loan fees and loan quantities. This is the key empirical advance relative to studies that observe only short interest, only lending fees, or only proxies for short-sale constraints.
- **Data advantage.** The authors use proprietary daily stock-lending data from a large institutional investor over September 1999–August 2003. The data contain the rebate rate, market rate, loan fee, the number of shares on loan, and the number of contracts for individual stocks.
- **Identification idea.** In a lending market, the loan fee is the price of borrowing shares and the amount on loan is the quantity. A rise in both price and quantity reveals at least an outward shift in shorting demand. A fall in price and rise in quantity reveals at least an outward shift in lending supply.
- **Main result.** Outward shorting demand shifts, denoted DOUT, strongly predict negative future stock returns. In the main monthly regression, DOUT predicts approximately -2.98% abnormal return over the next month.
- **Asymmetry.** The predictive power is concentrated in demand shifts out. Supply shifts and demand shifts in do not have comparable predictive power.
- **Information interpretation.** The negative returns after outward demand shifts are stronger when the public information environment is weaker. This supports the idea that short sellers trade on private information.
- **Portfolio implication.** Portfolios long demand-in stocks and short demand-out stocks earn positive abnormal returns, especially on an equal-weight basis.
- **Broader takeaway.** The shorting market is not merely a market for constraints. It is a market in which informed traders reveal negative information through their willingness to pay to borrow shares.

2 Incremental contribution of the paper

2.1 Relative to the short-sale constraint literature

- Earlier work showed that stocks with high shorting costs, high short interest, or limited lendable supply tend to have low subsequent returns.
- But these measures mix supply and demand. For example, a high lending fee could mean high shorting demand, low lendable supply, or both.
- This paper's incremental contribution is to observe both the price and quantity in the lending market and to infer the direction of supply and demand shifts.
- The distinction is economically essential: only outward demand shifts robustly predict negative returns.

Interpretation: The paper moves the literature from "short-sale constraints matter" to "the source of the constraint and the direction of the shift matter." This is a sharper test of the theory linking short sellers' information to future returns.

2.2 Relative to short-interest studies

- Short interest is an equilibrium quantity and cannot separately identify demand and supply.

- A rise in short interest may reflect more bearish demand, more lending supply, or both.
- Cohen, Diether, and Malloy show that changes in quantity alone do not have the same predictive power as demand-out classifications based on both fees and quantities.

Interpretation: The paper explains why short-interest results can be noisy: the same quantity change can mean different things depending on the lending fee.

2.3 Relative to asset-pricing and market-efficiency research

- The paper provides a mechanism for how negative private information enters prices.
- Short sellers reveal bad news not only by shorting shares, but by bidding up borrowing costs when the desire to short increases.
- The post-shift return predictability shows that the price discovery process is gradual: prices do not immediately incorporate all information embedded in the stock-lending market.

Interpretation: The contribution is to connect a microstructure market–the stock-loan market–to cross-sectional return predictability.

2.4 Relative to empirical identification in finance

- The paper adapts the textbook price-quantity logic of supply and demand to a financial-market setting.
- The empirical design does not require estimating full supply and demand curves.
- It only uses sign restrictions: fee up and quantity up implies demand shifted outward; fee down and quantity up implies supply shifted outward.

Interpretation: This is a clean and intuitive identification strategy. It avoids the much stronger assumption that either demand or supply is fixed.

3 Data and measurement

3.1 Stock-lending market data

- The data come from a large institutional investor that lends shares to short sellers through the stock-loan market.
- The sample covers September 1999 to August 2003.
- The sample includes NYSE, AMEX, and Nasdaq stocks with market capitalization below the NYSE median and stock prices above $5.
- The focus on smaller stocks is useful because lending constraints, information asymmetry, and shorting frictions are more likely to matter.

Interpretation: The data observe actual loan market transactions and thus provide direct measures of both shorting cost and quantity borrowed.

3.2 Loan fee

The paper defines the loan fee as

$$\mathrm{Fee}_{i,t} = MarketRate_t - RebateRate_{i,t}. \tag{1}$$

- $MarketRate_t$ is the relevant market interest rate.
- $RebateRate_{i,t}$ is the interest rate paid by the lender to the short seller on cash collateral.
- A higher fee means it is more costly to borrow shares.

Interpretation: The loan fee is the price in the shorting market. It is analogous to the rental price of shares.

3.3 Quantity on loan

The paper measures lending quantity as

$$\mathrm{Q}_{i,t} = \frac{SharesOnLoan_{i,t}}{SharesOutstanding_{i,t}}. \tag{2}$$

- This is the fraction of a firm's shares outstanding that are currently on loan from the lender.
- The paper also observes the number of active lending contracts.
- In robustness tests, the authors use aggregate short interest as an alternative quantity measure.

Interpretation: The amount on loan is the quantity in the lending-market supply-demand diagram. Quantity is important because a high fee alone does not reveal whether demand increased or supply decreased.

3.4 Shift classification

At the last trading day of month t, the paper compares the loan fee and quantity on loan to the previous month. It classifies stocks into four shift types:

$$\mathrm{DOUT}_{i,t} = 1\{\Delta\mathrm{Fee}_{i,t} > 0,\ \Delta\mathrm{Q}_{i,t} > 0\}, \tag{3}$$
$$\mathrm{DIN}_{i,t} = 1\{\Delta\mathrm{Fee}_{i,t} < 0,\ \Delta\mathrm{Q}_{i,t} < 0\}, \tag{4}$$
$$\mathrm{SOUT}_{i,t} = 1\{\Delta\mathrm{Fee}_{i,t} < 0,\ \Delta\mathrm{Q}_{i,t} > 0\}, \tag{5}$$
$$\mathrm{SIN}_{i,t} = 1\{\Delta\mathrm{Fee}_{i,t} > 0,\ \Delta\mathrm{Q}_{i,t} < 0\}. \tag{6}$$

- DOUT means demand for shorting shifted outward: price and quantity both rise.
- DIN means demand shifted inward: price and quantity both fall.
- SOUT means supply of lendable shares shifted outward: price falls and quantity rises.
- SIN means supply shifted inward: price rises and quantity falls.

Interpretation: These classifications identify at least one required shift. For example, a simultaneous increase in fee and quantity cannot be explained by a pure supply shift outward; demand must have moved outward.

3.5 Returns and controls

- The dependent variable is the next-month abnormal stock return.
- The baseline abnormal return is characteristically adjusted using 25 size/book-to-market portfolios.
- Robustness checks also use 75 size/book-to-market/momentum portfolios.
- Controls include last month's return, returns from months $t-12$ to $t-2$, institutional ownership, volume, and calendar month dummies.

Interpretation: The goal is to ask whether lending-market shifts predict returns after controlling for standard cross-sectional return predictors.

4 Empirical specifications

4.1 Main cross-sectional return regression

The main monthly regression is

$$AR_{i,t+1} = \alpha_t + eta_1 \mathrm{DIN}_{i,t} + \beta_2 \mathrm{DOUT}_{i,t} + \beta_3 \mathrm{SIN}_{i,t} + \beta_4 \mathrm{SOUT}_{i,t} + \gamma' X_{i,t} + \epsilon_{i,t+1}. \tag{7}$$

- $AR_{i,t+1}$ is the abnormal return in month $t+1$.
- $X_{i,t}$ contains lagged returns, institutional ownership, volume, and other controls.
- Calendar month dummies absorb common time effects.
- Standard errors are clustered by calendar date.

Interpretation: The coefficient β_2 on DOUT is the central object. It measures the return predictability of outward shorting demand shifts.

4.2 Fee and quantity interactions

The paper tests whether the magnitude of fee and quantity changes matters:

$$AR_{i,t+1} = \alpha_t + \beta_1 \mathrm{DOUT}_{i,t} + \beta_2 \Delta\mathrm{Fee}_{i,t} + \beta_3 \Delta\mathrm{Q}_{i,t} + \beta_4 (\Delta\mathrm{Fee}_{i,t} \times \Delta\mathrm{Q}_{i,t}) + \gamma' X_{i,t} + \epsilon_{i,t+1}. \tag{8}$$

Interpretation: If demand shifts reveal private information, large increases in both borrowing cost and borrowing quantity should be especially informative.

4.3 Portfolio tests

The paper forms monthly portfolios based on the shift categories:

$$R_{t+1}^{DIN-DOUT} = R_{t+1}^{DIN} - R_{t+1}^{DOUT}. \tag{9}$$

It reports raw excess returns and characteristic-adjusted abnormal returns.

Interpretation: The long-short portfolio test translates the regression result into an implementable trading strategy, though actual implementation must account for borrowing costs and shorting risks.

4.4 Information environment tests

The paper studies whether public information flow weakens the predictive power of DOUT:

$$AR_{i,t+1} = \alpha_t + \beta_1 \mathrm{DOUT}_{i,t} + \beta_2 PublicInfo_{i,t} + \beta_3 (\mathrm{DOUT}_{i,t} \times PublicInfo_{i,t}) + \gamma' X_{i,t} + \epsilon_{i,t+1}. \tag{10}$$

Public information proxies include residual analyst coverage and dividend payment status.

Interpretation: If DOUT captures private information, the effect should be strongest when public information is scarce.

5 Identification and empirical design

5.1 What identifies demand and supply shifts

- The paper treats the stock-lending market as a market with a price and quantity.
- The loan fee is the price.
- The fraction of shares on loan is the quantity.
- The signs of price and quantity movements reveal whether shorting demand or lendable supply must have shifted.

Interpretation: This is a revealed-shift strategy. It does not require instruments for demand or supply, only observed price-quantity movements.

5.2 Why the design improves on earlier measures

- Short interest measures only quantity.
- Loan fee measures only price.
- The joint movement of price and quantity distinguishes demand changes from supply changes.
- This distinction matters because only demand-out events predict low future returns.

Interpretation: The paper shows that the same level of shorting activity can have different implications depending on why the activity changed.

5.3 Information versus constraint interpretation

- A pure short-sale constraint story says high borrowing costs should predict overpricing.
- The paper finds that loan fees and quantities alone do not explain returns as strongly as DOUT.
- This suggests the relevant signal is new bearish demand, not merely a high level of constraints.
- The result is stronger when public information is limited, consistent with privately informed short sellers.

Interpretation: The paper's preferred interpretation is that short sellers use the lending market to act on negative private information.

5.4 Limits of the design

- The proprietary lender is large but not the entire lending market.
- Quantity on loan is lender-specific, though the paper checks aggregate short interest in robustness tests.
- The shift classification uses monthly changes and cannot observe every intramonth motive.
- Trading strategies based on the results must consider borrowing costs, recall risk, and market impact.

Interpretation: The evidence is strong for the informational role of demand shifts, but it does not imply every DOUT event is caused by informed trading.

6 Tables and figures

6.1 Table I and Table II: lending data and shift classification

Read:

- Table I illustrates how the loan fee equals the market rate minus the rebate rate.
- The sample has meaningful variation in fees and loan quantities, especially among below-median-size firms.
- Table II shows the shift categories. DOUT stocks experience loan fees rising from 3.30 to 3.72 and quantity on loan rising from 0.78 to 1.10.
- Demand-out stocks are relatively small and have high book-to-market and high volatility rankings.

Economic message: The data show that the lending market contains both price and quantity variation, which enables the supply-demand identification strategy.

Table I

Lending Activity Examples and Sample Summary Statistics

Panel A reports lending activity examples from our sample of proprietary stock lending data on a single date (August 29, 2003). For a given stock-day observation we use the rebate rate of the largest short sale contract (largest = most shares on loan). Market Rate refers to the collateral account interest rate. The loan fee (*Fee*) is the difference between the market rate and the rebate rate, and is the interest rate the lender receives from the short sale. *%On Loan* is the total number of shares on loan by our lender expressed as a percentage of total shares outstanding. *Num Cont* is the number of short sale lending contracts that the lender is engaged in for a given stock-day observation. Ptile *ME* is the NYSE market cap percentile. Panel B reports summary statistics for our entire sample (September 1999 to August 2003).

Panel A: Lending Activity Examples (August 29, 2003)

Stock	Rebate Rate	Market Rate	Fee	%On Loan	Num Cont	Ptile *ME*
Intel	0.95	1.00	0.05	0.01	1	99.8
Johnson & Johnson	0.95	1.00	0.05	0.03	2	99.7
PeopleSoft	0.00	1.00	1.00	0.00	1	88.0
Bally Total Fitness	0.25	1.00	0.75	1.78	14	33.0
American Superconductor	−1.50	1.00	2.50	5.51	40	28.4
Atlas Air	−6.25	1.00	7.25	4.75	26	4.5
Questcor Pharmaceutical	−13.75	1.00	14.75	0.34	10	3.9

Panel B: Lending Sample Summary Statistics

	Mean	Median	25 Ptile	75 Ptile
		All Stocks		
Rebate rate	0.55	0.25	−0.00	1.10
Market rate	3.15	1.94	1.37	5.22
Fee	2.60	1.82	0.14	4.20
%On loan	0.58	0.16	0.03	0.50
Num cont	9.09	4.00	2.00	8.00
Ptile *ME*	38	28	7	62
		Above NYSE Median *ME*		
Rebate rate	1.75	1.53	1.09	1.64
Market rate	2.14	1.69	1.23	1.92
Fee	0.39	0.13	0.10	0.16
%On Loan	0.14	0.03	0.001	0.08
Num cont	4.64	3	1	4
Ptile *ME*	78	81	65	89
		Below NYSE Median *ME*		
Rebate rate	−0.17	0.00	−0.01	0.12
Market rate	3.77	3.86	1.72	5.62
Fee	3.94	3.93	1.99	5.30
%On Loan	0.85	0.38	0.10	0.86
Num cont	11.79	6	2	11
Ptile *ME*	15	10	4	20

Table I: Lending activity and sample summary

Table II

Supply and Demand Shifts: Summary Statistics

This table reports summary statistics for shifts in shorting supply and shorting demand from the universe of NYSE, AMEX, and Nasdaq stocks with lagged market capitalization below the NYSE median and lagged price greater than or equal to \$5. Panel A presents means, and Panel B presents medians. Shifts are constructed as follows. The last trading day of month t we check if there was a shift in shorting supply or shorting demand during the month (based on changes in loan fees and changes in the percentage of shares lent out). We place stocks into shift categories: demand in (*DIN*), demand out (*DOUT*), supply in (*SIN*), and supply out (*SOUT*). *Before Shift Loan Fee* is the lending fee before the shift. *New Loan Fee* is the lending fee when the shift occurs. *Before Shift %On Loan* is the number of shares on loan by our lender before the shift occurs as a percentage of shares outstanding. *New %On Loan* is the percentage of shares on loan by our lender when the shift occurs. *ME* is market cap, and *BE/ME* is the book-to-market ratio. *Vol* is the average daily exchange-adjusted turnover of a stock during the past 6 months. The time period is September 1999 to August 2003.

	DIN	*DOUT*	*SIN*	*SOUT*
		Panel A: Mean		
Number of stocks per month	34	22	38	31
Percentile *ME*	25	22	22	23
Percentile *BE/ME*	37	32	38	34
Percentile Vol	72	74	71	74
Before Shift Loan Fee	2.57	3.30	2.88	3.11
New Loan Fee	2.16	3.72	3.28	2.55
Before Shift %On Loan	1.09	0.78	0.98	0.79
New %On Loan	0.80	1.10	0.65	1.09
		Panel B: Median		
Number of stocks per month	34	14	21	34
Percentile *ME*	22	19	19	22
Percentile *BE/ME*	29	23	29	24
Percentile Vol	79	81	79	82
Before Shift Loan Fee	2.07	3.52	2.47	2.69
New Loan Fee	1.74	4.22	3.05	2.07
Before Shift %On Loan	0.58	0.40	0.58	0.34
New %On Loan	0.35	0.63	0.30	0.62

Table II: Supply and demand shift summary

6.2 Table III and Figure 1: main return predictability and dynamics

Read:

- Table III shows that DOUT is strongly negatively related to next-month abnormal returns.
- The main coefficient is about −2.98% with a large t-statistic.
- DIN, SIN, and SOUT are not comparably significant.
- Figure 1 shows that daily, weekly, and monthly return patterns are most negative following lagged DOUT shifts.

Economic message: The shorting demand-out signal predicts gradual negative price adjustment, consistent with private information being incorporated over time.

Table III

Cross-Sectional Regressions: Monthly Abnormal Returns

The table reports estimates from pooled, cross-sectional regressions of the monthly abnormal returns (in percent) of all NYSE, AMEX, and Nasdaq stocks with market capitalization below the NYSE median with lagged share prices above $5 on supply and demand shift dummy variables and a host of control variables. We characteristically adjust the left-hand side returns for size and book-to-market using 25 equal-weight size/book-to-market portfolios. *DIN* is a dummy variable for an inward demand shift last month. *DOUT* is a dummy variable for an outward demand shift last month. *SIN* is a dummy variable for an inward supply shift last month. *SOUT* is a dummy variable for an outward supply shift last month. r_{-1} is last month's return. $r_{-12,-2}$ is the return from month $t-12$ to $t-2$. *IO* is institutional ownership measured as a fraction of shares outstanding lagged one quarter. *Volume* is the average daily exchange-adjusted share turnover during the previous 6 months. *Fee* > x equals one if the loan fee is greater than x and zero otherwise. *Quantity* is the fraction of shares outstanding on loan by the lender at the end of month $t-1$. The regressions include calendar month dummies, and the standard errors take into account clustering by calendar date. The time period is October 1999 to September 2003. t-statistics are in parentheses. The intercept is estimated but not reported.

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
DIN	0.281	0.299					0.502		−0.149
	(0.43)	(0.59)					(0.96)		(0.31)
DOUT	−3.144	−2.983					−2.364		−2.485
	(3.20)	(3.96)					(3.27)		(3.19)
SIN	0.006	−0.063					0.443		0.176
	(0.01)	(0.08)					(0.69)		(0.22)
SOUT	−0.869	−0.820					−0.569		−1.225
	(1.11)	(1.23)					(0.85)		(1.81)
r_{-1}		−0.012	−0.012	−0.012	−0.012	−0.012	−0.012	−0.013	−0.012
		(0.49)	(0.50)	(0.49)	(0.48)	(0.48)	(0.48)	(0.51)	(0.50)
$r_{-12,-2}$		0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005
		(0.83)	(0.84)	(0.84)	(0.84)	(0.83)	(0.82)	(0.85)	(0.82)
IO		0.309	0.320	0.317	0.276	0.278	0.255	0.419	0.317
		(0.44)	(0.48)	(0.47)	(0.41)	(0.40)	(0.37)	(0.59)	(0.45)
Volume		0.018	−0.004	0.001	0.004	0.006	0.019	−0.024	0.018
		(0.05)	(0.01)	(0.00)	(0.01)	(0.02)	(0.06)	(0.07)	(0.05)
Fee > 3.0%			−0.767	−0.557					
			(0.94)	(0.61)					
Fee > 5.0%					−2.025	−1.585	−1.596		
					(2.09)	(1.66)	(1.63)		
Quantity				−0.138		−0.044			
				(0.48)		(0.19)			
Quantity ∗ (*Fee* > 3.0%)				−0.134					
				(0.34)					
Quantity ∗ (*Fee* > 5.0%)						−0.582			
						(1.05)			
Δ*Fee*								−0.920	−0.912
								(1.95)	(1.76)
Δ*Quantity*								−0.907	−0.343
								(2.45)	(0.77)
Observations/month	2,100	2,098	2,098	2,098	2,098	2,098	2,098	2,098	2,098

Table III: Monthly abnormal return regressions

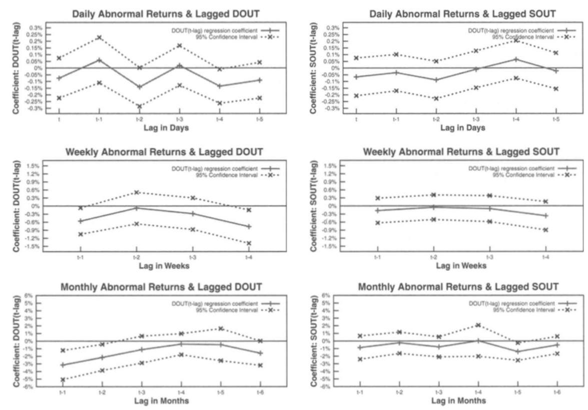

Figure 1. Daily, weekly, and monthly abnormal returns. In the top row we regress the daily abnormal returns (in percent) of all stocks owned by the lender with market capitalization below the NYSE median and lagged share prices above $5 on daily supply and demand shifts and lagged daily returns. In row 2 (row 3) we regress the weekly (monthly) abnormal returns (in %) of the same universe of stocks on weekly (monthly) supply and demand shifts. We include all the shifts in the regressions (*DIN*, *DOUT*, *SIN*, and *SOUT*), but only report the coefficients for *DOUT* and *SOUT*. *DOUT*($t - lag$) is a dummy variable for an outward demand shift lag days (weeks, months) ago, and *SOUT*($t - lag$) is a dummy variable for an outward supply shift lag days (weeks, months) ago. We proxy for expected returns characteristically using 25 equal-weight size/book-to-market portfolios. We run separate regressions for each lag length. The regressions include calendar time dummies, and the standard errors take into account clustering by calendar date.

Figure 1: Daily, weekly, and monthly return dynamics

6.3 Table IV and Table V: magnitude and portfolio returns

Read:

- Table IV shows that the DOUT effect is stronger when changes in fees and quantities are large.
- Interactions involving large fee and quantity changes show especially negative returns.
- Table V reports shift-based portfolio returns.
- The long DIN / short DOUT portfolio earns positive abnormal returns, particularly for equal-weight portfolios.

Economic message: The magnitude of the price-quantity movement matters. The more intense the outward shorting demand shift, the stronger the subsequent negative return.

Table IV

Cross-Sectional Regressions: Large Shifts and High Loan Fees

The table reports estimates from pooled, cross-sectional regressions of the monthly abnormal returns (in percent) of all NYSE, AMEX, and Nasdaq stocks with market capitalization below the NYSE median with lagged share prices above $5 on supply and demand shifts and control variables. We characteristically adjust the left-hand side returns for size and book-to-market using 25 equal-weight size/book-to-market portfolios. We also interact the supply and demand shifts with loan fee and quantity changes. *DIN* (*DOUT*) is a dummy variable for an inward (outward) demand shift last month, and *SIN* (*SOUT*) a dummy variable for an inward (outward) supply shift last month. $Fee > 3\%$ is a dummy variable that equals one if the loan fee is greater than 3%. ΔFee^{+}_{big}(ΔFee^{-}_{big}) is a dummy variable that equals one if the change in the loan fee for month $t-1$ is greater than (less than or equal to) the 90th (10th) percentile. $\Delta Quantity^{+}_{big}$ is a dummy variable that equals one if the change in quantity on loan (as a percentage of shares outstanding) for month $t-1$ is greater than the 90th percentile. r_{-1} is last month's return. $r_{-12,-2}$ is the return from month $t-12$ to $t-2$. *IO* is institutional ownership as a fraction of shares outstanding lagged one quarter. *Volume* is the average daily exchange-adjusted share turnover during the previous 6 months. All regressions include calendar month dummies, and all standard errors take into account clustering by calendar date. The time period is October 1999 to September 2003. *t*-statistics are in parentheses. The intercept is estimated but not reported.

	[1]	[2]	[3]	[4]	[5]	[6]
DIN	0.299	0.297	0.298	0.294	0.365	−0.613
	(0.59)	(0.58)	(0.59)	(0.58)	(0.69)	(1.01)
DOUT	−2.983	−2.649	−2.649	−2.458	−2.866	−0.769
	(3.96)	(2.66)	(2.81)	(2.95)	(3.40)	(0.97)
SIN	−0.063	−0.062	−0.063	−0.059	0.038	0.655
	(0.08)	(0.08)	(0.08)	(0.07)	(0.05)	(0.88)
SOUT	−0.820	−0.819	−1.175	−1.136	−0.744	−0.624
	(1.23)	(0.98)	(1.53)	(1.53)	(1.17)	(0.95)
$DOUT \times \Delta Fee^{+}_{big}$		−1.253				
		(0.69)				
$SOUT \times \Delta Fee^{-}_{big}$		−0.015				
		(0.01)				
$DOUT \times \Delta Quantity^{+}_{big}$			−1.138			
			(0.67)			
$SOUT \times \Delta Quantity^{+}_{big}$			1.169			
			(1.02)			
$\Delta Fee^{+}_{big} \times \Delta Quantity^{+}_{big}$				−4.480		
				(2.32)		
$\Delta Fee^{-}_{big} \times \Delta Quantity^{+}_{big}$				2.633		
				(1.32)		
$Fee > 3\%$					−0.219	0.175
					(0.24)	(0.24)
$(Fee > 3\%) \times DIN$						2.798
						(1.82)
$(Fee > 3\%) \times DOUT$						−4.144
						(2.69)
$(Fee > 3\%) \times SIN$						−1.691
						(0.93)
$(Fee > 3\%) \times SOUT$						−0.710
						(0.42)
Observations/month	2,098	2,098	2,098	2,098	2,098	2,098
Control Variables	$r_{-1}, r_{-12,-2}$, *IO*, *Volume*, and calendar month dummies					

Table IV: Large shifts and high loan fees

Table V

Supply and Demand Shifts: Monthly Portfolio Returns (in percent)

This table presents average monthly returns (in percent) on shorting supply and shorting demand shift portfolios from the universe of NYSE, AMEX, and Nasdaq stocks with lagged market capitalization below the NYSE median and lagged price greater than or equal to $5. Excess Returns are average monthly returns in excess of the 1-month Treasury bill rate. Abnormal Returns are computed by characteristically adjusting returns using 25 equal weight size/book-to-market portfolios and 75 ($3 \times 5 \times 5$) equal weight size/book-to-market/momentum portfolios. The benchmark portfolios also contain the restriction that lagged price must be greater than or equal to $5. For the last trading day of month $t-1$ we check if there was a shift in shorting supply or shorting demand during the month (based on changes in loan fees and changes in the percentage of shares lent out). We place stocks into shift portfolios: demand in (*DIN*), demand out (*DOUT*), supply in (*SIN*), and supply out (*SOUT*). Shift portfolios are formed in month $t-1$ and the stocks are held in the portfolios during month t. The time period is October 1999 to September 2003.

	DIN	*DOUT*	*SIN*	*SOUT*	*DIN − DOUT*	*SIN − SOUT*
Panel A: Excess Returns						
Equal-weight						
Mean	1.65	−1.82	0.84	−1.12	3.48	1.96
t-stat	0.79	−1.02	0.49	−0.57	2.34	1.55
Value-weight						
Mean	0.53	−0.53	0.42	−2.15	1.06	2.57
t-stat	0.27	−0.27	0.23	−1.09	0.65	1.92
Panel B: Abnormal Returns (Benchmark Portfolios: 25 Size/Book-to-Market Portfolios)						
Equal-weight						
Mean	0.84	−2.34	0.48	−1.81	3.18	2.28
t-stat	0.67	−2.52	0.56	−1.43	2.17	1.81
Value-weight						
Mean	−0.12	−1.05	0.01	−2.63	0.93	2.65
t-stat	−0.12	−0.88	0.01	−2.23	0.56	1.94
Panel C: Abnormal Returns (Benchmark Portfolios: 75 Size/Book-to-Market/Momentum Portfolios)						
Equal-weight						
Mean	0.76	−2.11	0.08	−1.63	2.87	1.72
t-stat	0.72	−2.15	0.11	−1.36	2.21	1.34
Value-weight						
Mean	−0.26	−0.88	−0.29	−2.41	0.61	2.12
t-stat	−0.28	−0.75	−0.30	−2.15	0.41	1.58

Table V: Monthly portfolio returns

6.4 Table VI and Table VII: robustness and information environment

Read:

- Table VI shows that the DOUT effect survives industry controls, alternative quantity measures, and restrictions based on the lender's market power.
- The interaction with lender market power suggests stronger results when the lender accounts for a larger fraction of lending activity.
- Table VII tests whether public information proxies attenuate the DOUT effect.
- Evidence using analyst coverage and dividends supports the view that the effect is tied to private information rather than public-news trading.

Economic message: The demand-out result is not a fragile artifact of a particular specification. It is consistent with the idea that short sellers reveal private negative information.

Table VI

Cross-Sectional Regressions: Robustness Tests

The table reports estimates from pooled, cross-sectional regressions of the monthly abnormal returns (in percent) of all NYSE, AMEX, and Nasdaq stocks owned by the lender with market capitalization below the NYSE median and lagged share prices above $5 on supply and demand shift dummy variables and a host of control variables. We characteristically adjust all left-hand side returns for size and book-to-market using 25 equal-weight size-*BE/ME* portfolios. *DIN* (*DOUT*) is a dummy variable for an inward (outward) demand shift last month, and *SIN* (*SOUT*) a dummy variable for an inward (outward) supply shift last month. In columns 1 to 4, the loan quantity used to define shifts is the number of shares on loan by our lender divided by total shares outstanding (*%On Loan*), and returns are measured at the month's end; in column 5, loan quantity equals total monthly short interest divided by total shares outstanding (*Short Int*), and monthly returns are measured using closing prices from the 16th of the month. *Market Power* is the number of shares lent out by our lender in month $t-1$ divided by short interest (*SI*) in month $t-1$, while *Market Power* > 2/3 is a dummy variable equal to one if the lender's Market Power exceeds two-thirds. r_{-1} is last month's return. $r_{-12,-2}$ is the return from month $t-12$ to $t-2$. *IO* is institutional ownership as a fraction of shares outstanding lagged one quarter. *Volume* is the average daily exchange-adjusted share turnover during the previous six months. All five columns include calendar month dummies, and the first column also includes industry dummies (using Fama and French's (1997) 48-industry classification scheme). All standard errors take into account clustering by calendar date. The time period is October 1999 to September 2003 for columns 1 to 4 and November 1999 to September 2003 for column 5. t-statistics are in parentheses. The intercept is estimated but not reported.

	[1]	[2]	[3]	[4]	[5]
DIN	0.206	0.273	0.196	0.356	0.215
	(0.41)	(0.53)	(0.34)	(0.69)	(0.42)
DOUT	−3.030	−3.014	−2.389	−2.741	−1.262
	(4.05)	(3.94)	(3.03)	(3.52)	(2.06)
SIN	−0.172	−0.067	0.156	0.009	−0.950
	(0.22)	(0.08)	(0.18)	(0.01)	(1.22)
SOUT	−0.888	−0.804	−0.731	−0.803	−0.793
	(1.35)	(1.21)	(1.17)	(1.25)	(1.48)
(*Market Power*) × *DIN*			0.585		
			(0.25)		
(*Market Power*) × *DOUT*			−5.642		
			(1.26)		
(*Market Power*) × *SIN*			−1.537		
			(0.56)		
(*Market Power*) × *SOUT*			−0.683		
			(0.24)		
(*Market Power* > 2/3) × *DIN*				−1.446	
				(0.77)	
(*Market Power* > 2/3) × *DOUT*				−8.361	
				(2.38)	
(*Market Power* > 2/3) × *SIN*				−1.296	
				(0.57)	
(*Market Power* > 2/3) × *SOUT*				0.050	
				(0.01)	
Observations/month	2,098	2,095	2,095	2,095	2,097
Loan Quantity	*%On Loan*	*%On Loan*	*%On Loan*	*%On Loan*	*Short Int*
Industry Dummies	Yes	No	No	No	No
Control Variables	$r_{-1}, r_{-12,-2}$, *IO*, *Volume*, and calendar month dummies always included				

Table VI: Robustness tests

Table VII

Cross-Sectional Regressions: Analyst Coverage and Predictable Demand

The table reports estimates from pooled, cross-sectional regressions of the monthly abnormal returns (in percent) of all NYSE, AMEX, and Nasdaq stocks owned by the lending institution with market capitalization below the NYSE median and lagged share prices above $5 on supply and demand shift dummy variables, residual analyst coverage, a dividend indicator variable, and a host of other control variables. We characteristically adjust the left-hand side returns for size and book-to-market using 25 equal-weight size/book-to-market portfolios. *DIN* (*DOUT*) is a dummy variable for an inward (outward) demand shift last month, and *SIN* (*SOUT*) a dummy variable for an inward (outward) supply shift last month. *RCOV* is last month's residual analyst coverage, and is computed by running a cross-sectional regression of analyst coverage on size, and then calculating the residual for each stock. This residual is defined as residual analyst coverage. *DIV* is a dummy variable equal to one if there was a dividend in month $t-1$. The control variables r_{-1} is last month's return. $r_{-12,-2}$ is the return from month $t-12$ to $t-2$. *IO* is institutional ownership as a fraction of shares outstanding lagged one quarter. *Volume* is the average daily exchange-adjusted share turnover during the previous six months. All regressions include calendar month dummies, and all standard errors take into account clustering by calendar date. The time period is October 1999 to September 2003. t-statistics are in parentheses. The intercept is estimated but not reported.

	[1]	[2]	[3]
DIN	0.299	0.303	0.285
	(0.59)	(0.60)	(0.56)
DOUT	−2.983	−2.936	−3.096
	(3.96)	(4.30)	(3.91)
SIN	−0.063	−0.059	−0.075
	(0.08)	(0.08)	(0.10)
SOUT	−0.820	−0.813	−0.835
	(1.23)	(1.22)	(1.26)
RCOV		0.019	
		(0.26)	
RCOV × *DOUT*		0.052	
		(0.24)	
DIV			−0.347
			(1.09)
DIV × *DOUT*			2.251
			(0.94)
Observations/month	2,098	2,098	2,098
Control variables	$r_{-1}, r_{-12,-2}$, *IO*, *Volume*, and calendar month dummies		

Table VII: Analyst coverage and predictable demand

6.5 Table VIII: indirect costs and risks of shorting

Read:

- Table VIII examines whether the DOUT effect is explained by trading volume, firm size, or idiosyncratic risk.
- The effect remains negative after interacting DOUT with proxies for shorting risks and costs.
- The DOUT effect is especially strong for small stocks, where short-sale constraints and information asymmetry are more likely to bind.

Economic message: Borrowing costs and shorting risks matter, but they do not eliminate the core information content of outward shorting demand shifts.

Table VIII

Cross-Sectional Regressions: Indirect Costs and Risks of Shorting

The table reports estimates from pooled, cross-sectional regressions of the monthly abnormal returns (in percent) of all NYSE, AMEX, and Nasdaq stocks owned by the lender with market capitalization below the NYSE median and lagged share prices above five dollars on supply and demand shift dummy variables, proxies for indirect risks and costs of shorting, and control variables. We characteristically adjust the left-hand side returns for size and book-to-market using 25 equal-weight size/book-to-market portfolios. *DIN* (*DOUT*) is a dummy variable for an inward (outward) demand shift last month, and *SIN* (*SOUT*) a dummy variable for an inward (outward) supply shift last month. r_{-1} is last month's return. $r_{-12,-2}$ is the return from month $t-12$ to $t-2$. *IO* is institutional ownership as a fraction of shares outstanding lagged one quarter. Volume is the average daily exchange-adjusted share turnover during the previous 6 months. $Volume_{high}$ is a dummy variable that equals one if the average daily exchange-adjusted share turnover during the previous 6 months is greater than the 80th volume percentile of all stocks in the regression. ME_{small} is a dummy variable that equals one if lagged market-cap is less than or equal to the 20th lagged market-cap percentile of all stocks in the regression. *IRISK* is idiosyncratic risk measured as the standard deviation of market model residuals using 1 year of past daily returns. The market model regression is $r_{it} = \alpha_i + \beta_{1i} r_{Mt} + \beta_{2i} r_{Mt-1} + \varepsilon_{it}$ where r_{it} is the return on stock i for day t and r_{Mt} is the return on the market portfolio (CRSP value-weight index) for day t. $IRISK_{high}$ is a dummy variable that equals one if IRISK is greater than the 80th *IRISK* percentile of all stocks in the regression. All regressions include calendar month dummies, and all standard errors take into account clustering by calendar date. The time period is October 1999 to September 2003. t-statistics are in parentheses. The intercept is estimated but not reported.

	[1]	[2]	[3]	[4]
DIN	0.299	0.266	0.297	0.284
	(0.59)	(0.53)	(0.58)	(0.55)
DOUT	−2.983	−3.015	−2.407	−2.155
	(3.96)	(3.00)	(2.90)	(3.15)
SIN	−0.063	−0.088	−0.063	−0.081
	(0.08)	(0.11)	(0.08)	(0.10)
SOUT	−0.820	−0.869	−0.824	−0.839
	(1.23)	(1.30)	(1.23)	(1.24)
$Volume_{high}$		0.500		
		(1.00)		
$Volume_{high} \times DOUT$		−0.001		
		(0.00)		
ME_{small}			−0.126	
			(0.48)	
$ME_{small} \times DOUT$			−3.903	
			(2.38)	
$IRISK_{high}$				0.111
				(0.13)
$IRISK_{high} \times DOUT$				−1.766
				(1.04)
Observations/month	2,098	2,098	2,098	2,098
Control variables	$r_{-1}, r_{-12,-2}$, *IO*, *Volume*, and calendar month dummies			

7 Short summary

- The paper uses proprietary daily stock-lending data from a large institutional investor.
- It observes both loan fees and quantities on loan.
- The joint movement of fee and quantity identifies demand and supply shifts in the shorting market.
- Outward shorting demand shifts predict negative future returns of roughly 3% over the following month.
- Other shifts do not have similar predictive power.
- The result is robust to controls for lagged returns, institutional ownership, volume, industry effects, and alternative quantity measures.

- Portfolio tests show positive returns to long DIN and short DOUT strategies before trading-cost adjustments.
- The effect is stronger when public information is scarce, supporting a private information interpretation.
- The paper's incremental contribution is to separate shorting demand from lendable supply and show that demand shifts are the relevant information signal.

8 Conclusion

8.1 Core conclusion

Cohen, Diether, and Malloy show that the stock-lending market contains strong information about future returns. The crucial signal is not simply high short interest or high borrowing cost; it is an outward demand shift for shorting, identified by simultaneous increases in borrowing fees and quantities on loan.

8.2 Economic intuition

Short sellers who receive negative information want to borrow and sell shares. When many such traders demand the same stock, both the price of borrowing and the quantity borrowed rise. Because the stock price does not immediately incorporate all of this information, demand-out shifts predict subsequent negative returns.

8.3 Takeaway

The paper turns the stock-lending market into a source of identified supply and demand shocks. Its broader message is that financial-market quantities are hard to interpret without prices. Observing both loan fees and loan quantities reveals which side of the shorting market moved and why that movement matters for returns.